

21-2a Representing Preferences with Indifference Curves

The consumer's preferences allow her to choose among different bundles of pizza and Pepsi. If you offer the consumer two different bundles, she chooses the bundle that best suits her tastes. If the two bundles suit her tastes equally well, we say that the consumer is *indifferent* between the two bundles.

Just as we have represented the consumer's budget constraint graphically, we can also represent her preferences graphically. We do this with indifference curves. An

indifference curve (a curve that shows consumption bundles that give the consumer the same level of satisfaction)

shows the various bundles of consumption that make the consumer equally happy. In this case, the indifference curves show the combinations of pizza and Pepsi with which the consumer is equally satisfied.

Figure 3 shows two of the consumer's many indifference curves. We can see that the consumer is indifferent among combinations A, B, and C because they are all on the same curve. Not surprisingly, if the consumer's consumption of pizza decreases, say, from point A to point B, consumption of Pepsi must increase to keep her equally happy. If consumption of pizza decreases again, from point B to point C, the amount of Pepsi consumed must increase yet again.

Figure 3

The Consumer's Preferences

The consumer's preferences are represented with indifference curves, which show the combinations of pizza and Pepsi that make the consumer equally satisfied.

Because the consumer prefers more of a good, points on a higher indifference curve (I_2) are preferred to points on a lower indifference curve (I_1). The marginal rate of substitution (MRS) shows the rate at which the consumer is willing to trade Pepsi for pizza. It measures the quantity of Pepsi the consumer must receive in exchange for 1 pizza.

The slope at any point on an indifference curve equals the rate at which the consumer is willing to substitute one good for the other. (The slope is negative, but for our purposes, we can ignore the minus sign.) This rate is called the marginal rate of substitution (the rate at which a consumer is willing to trade one good for another)

(*MRS*). In this case, the marginal rate of substitution measures how much additional Pepsi the consumer requires to be compensated for a one-unit reduction in pizza consumption. Notice that because the indifference curves are not straight lines, the marginal rate of substitution is not the same at all points on a given indifference curve. The rate at which a consumer is willing to trade one good for the other depends on the amounts of the goods she is already consuming. In other words, the rate at which a consumer is willing to trade pizza for Pepsi depends on whether she is hungrier or thirstier, and her hunger and thirst in turn depend on her current consumption of pizza and Pepsi.

The consumer is equally happy at all points on any given indifference curve, but she prefers some indifference curves to others. Because she prefers more consumption to less, higher indifference curves are preferred to lower ones. In Figure 3, any point on curve I_2 is preferred to any point on curve I_1 .

A consumer's set of indifference curves gives a complete ranking of the consumer's preferences. That is, we can use the indifference curves to rank any two bundles of goods. For example, the indifference curves tell us that the bundle at point D is preferred to the bundle at point A because point D is on a higher indifference curve than point A. (That conclusion may be obvious, however, because point D offers the consumer both more pizza and more Pepsi.) The indifference curves also tell us that the bundle at point D is preferred to the bundle at point C because point D is on a higher indifference curve. Even though point D has less Pepsi than point C, it has more than enough extra pizza to make the consumer prefer it. By seeing which point is on the higher indifference curve, we can use the set of indifference curves to rank any combination of pizza and Pepsi.

